

M.Sc. Assignment

Developing an LSTM System for Frequency Extraction from a Mixed Signal

1. Background and Goal

1.1 Problem Statement

A mixed signal composed of several sinusoidal waves (f_1-f_4) with added noise is given.
The goal is to develop a **Long Short-Term Memory (LSTM)** system capable of isolating and identifying the individual pure sinusoidal components that make up the noisy mixture.

1.2 The Principle

This task is formulated as a **conditional regression** problem:
the system receives the mixed signal $S(t)$ and a one-hot encoded vector C representing the target frequency to extract, and it outputs the reconstructed clean sinusoid corresponding to that frequency.

Input Output

$S[t], C \text{ Target}_i[t] = \sin(2\pi \cdot f_i \cdot t)$

1.3 Usage Example

If $C = [0 \ 1 \ 0 \ 0]$ corresponds to frequency f_2 , then

Input: $(S[t], C) \rightarrow \text{LSTM} \rightarrow \text{Output: Sinus_pure}_2[t]$

For example:

$S[0] + C \rightarrow \text{LSTM} \rightarrow \text{sinus_pure}_2[0]$

$S[1] + C \rightarrow \text{LSTM} \rightarrow \text{sinus_pure}_2[1]$

2. Dataset Creation

2.1 General Parameters

Parameter	Value
$f_1 = 1 \text{ Hz}, f_2 = 3 \text{ Hz}, f_3 = 5 \text{ Hz}, f_4 = 7 \text{ Hz}$	
Time range (t)	0 – 10 seconds
Sampling rate F_s	1000 Hz
Total samples	10 000 samples

2.2 Noisy Signal Creation (S)

Each sinusoid i is generated as:

$A_i(t) \sim \text{Uniform}(0.8, 1.2)$

$\phi_i(t) \sim \text{Uniform}(0, 2\pi)$

$\text{Sinus_noisy}_i(t) = A_i(t) \cdot \sin(2\pi \cdot f_i \cdot t + \phi_i(t))$

The mixed signal is:

$$S(t) = \frac{1}{4} \sum_{i=1}^4 \text{Sinus_noisy}_i(t)$$

2.3 Ground Truth Targets (no noise)

For each frequency f_i :

$$\text{Target}_i(t) = \sin(2\pi \cdot f_i \cdot t)$$

2.4 Train vs Test Sets

- Training set uses Seed #1
- Testing set uses Seed #2 (different random amplitudes and phases)

3. Training Dataset Structure

Each training sample contains 10 000 rows $\times$ 4 frequencies.

Input format: $(S[t], C_1, C_2, C_3, C_4)$

Each C is a one-hot vector selecting which frequency the model should reconstruct.

4. Pedagogical Notes – Internal State and Sequence Length

4.1 The Internal State of LSTM

Each LSTM cell maintains a **hidden state** h_t and **cell state** c_t , which capture temporal dependencies between consecutive samples.

4.2 Critical Implementation Requirement ($L = 1$)

For this assignment, **sequence length $L = 1$** must be used.

That means the internal state is reset between samples:

the model does not carry memory from one time step to another.

If not reset, residual state information could bias the next prediction.

Implementation Mode	Description
Reset State Each Sample ($L = 1$)	Keeps samples independent — required for this task.
Sequential Mode ($L > 1$)	Allows temporal dependencies but adds complexity. Not required here.

4.3 Alternative and Justification

Using $L > 1$ (sliding window of e.g. 10 or 50 samples) could improve training efficiency and convergence by capturing short-term dependencies, but for this exercise $L = 1$ is chosen for simplicity and clarity of concept.

5. Performance Evaluation

5.1 Success Metrics

Mean Squared Error (MSE) is used to evaluate performance on both train and test sets:

$$[\text{MSE}_{\text{train}} = \frac{1}{40000} \sum_{j=1}^{40000} (\text{LSTM}(S_{\text{train}}[t], C) - \text{Target}[t])^2]$$

$$[\text{MSE}_{\text{test}} = \frac{1}{40000} \sum_{j=1}^{40000} (\text{LSTM}(S_{\text{test}}[t], C) - \text{Target}[t])^2]$$

A good model should show $\text{MSE}_{\text{test}} \approx \text{MSE}_{\text{train}} \rightarrow$ good generalization.

5.2 Recommended Graphs

1. Overlay plot for f_2 showing:
 - Target signal (no noise)
 - Noisy input signal $S(t)$
 - LSTM output
2. Separate plots for each frequency (f_1 – f_4) comparing Target vs Predicted signals.

6. Assignment Summary

You are required to:

1. Generate two datasets (train and test) of noisy mixed signals using the given parameters.
2. Build an LSTM model that receives $(S[t], C)$ and predicts $\text{Target}_i[t]$.
3. Ensure that the internal state is reset between samples (sequence length $L = 1$).
4. Evaluate performance using MSE and illustrate the results with graphs.

The goal is to demonstrate that the LSTM can learn to extract a specific frequency component from a noisy mixture when properly conditioned by the one-hot vector C .

7. Hebrew Original Table of Contents

(Provided in the document as reference to the Hebrew sections.)

Would you like me to also produce a **clean, formatted Word or PDF version** of this English translation (e.g., same layout as the original)?